High Returns, Low Investment: Public Capital and the Euler Equation Puzzle

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Abstract

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1 Introduction

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2 An Euler Equation Model Incorporating Public Capitals

I build on the intertemporal optimisation framework of Kremer et al. (2019). I first recall the model setup and present the Euler Equation Puzzle through the model. I then consider the extreme scenario where only public capitals exist and investment occurs only through taxation, showing that this generates a collective action problem and a government efficiency problem that dilutes the effective return to capital when state taxation capacity and efficiency is low. Finally, I will extend this analysis into a generalised model incorporating both public and private capital, allowing households to allocate investments between the two.

2.1 Baseline: Standard Euler Equation

Consider in a deterministic discrete-time intertemporal consumption model, households maximise their discounted lifetime utility over an infinite time horizon.

Optimisation Problem

Households face the following optimisation problem:

$$\max_{\{C_t\}} \sum_{t=0}^{\infty} \delta^t u(C_t) \tag{1}$$

where $0 < \delta < 1$ is the annual discount factor, C_t is the household consumption in period t, and $u(C_t)$ measures the utility from consuming C_t in period t. $u(C)$ is increasing and concave and follows Inada Conditions: $u'(C) > 0$, $u''(C) < 0$, $\lim_{C\to 0} u'(C) = \infty$, and $\lim_{C\to\infty} u'(C) = 0$.

Budget Constraint

$$x_{t+1} = f'(K_t)(x_t - C_t) \tag{2}$$

x_t is household assets ('cash on hand') in period t with x_0 given, C_t is consumption in period t, and $f'(K_t)$ is the gross return to capital. $f(K)$ is increasing and concave, following: $f'(K) > 0$, $f''(K) < 0$, $\lim_{K\to 0} f'(K) = \infty$, and $\lim_{K\to\infty} f'(K) = 1$ (long-run steady state).

Discrete Time Euler Equation

Solving for optimal intertemporal consumption path:

$$u'(C_t) = \delta f'(K_t) u'(C_{t+1}) \tag{3}$$

Assume constant-elasticity-of-substitution (CES) utility:

$$u(C) = \begin{cases} \frac{C^{1-\sigma}}{1-\sigma} & \text{if } \sigma \neq 1 \\ \ln(C) & \text{if } \sigma = 1 \end{cases}$$

This means that the elasticity of intertemporal substitution is constant and independent of C_t. Therefore, the consumption growth rate is:

$$\frac{C_{t+1}}{C_t} - 1 = (\delta f'(K_t))^{\frac{1}{\sigma}} - 1 \tag{4}$$

The Puzzle

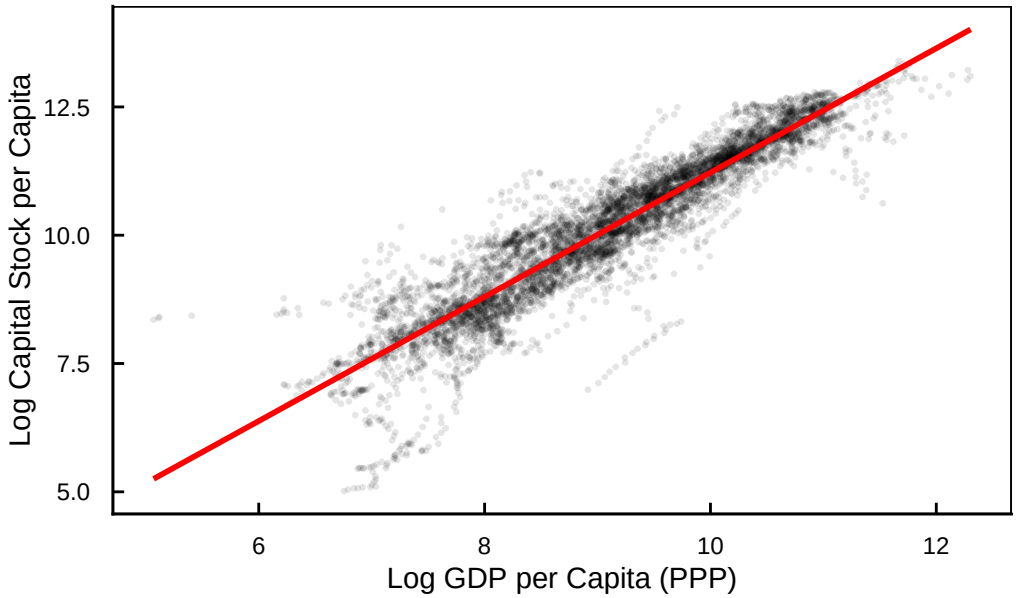
Assume that the model governs the intertemporal consumption behaviour of a representative household, the model should also apply at the macro level. To illustrate the puzzle with macro-level data, I use $\delta = 0.96$ [1]and $\sigma = 2$ [2] proposed in Kremer et al. (2019), and internal rates of return from the Penn World Table (PWT) [3] (Feenstra et al., 2015) as a proxy for $f'(K)$[4] to model the predicted consumption growth rates under Equation 4. Intuitively, a concave production function means that countries with lower capital stock face higher returns on capital investment. Figure 1 shows that countries with lower income have lower capital stock. Therefore, according to Equation 4, consumption growth rates should be high. However, Figure 2 Panel (A) shows that in many middle- and low-income countries, the actual growth rates fall below the model predictions. Moreover, Figure 2 Panel (B) shows that countries with higher predicted consumption growth rates tend to have larger gap between predicted and actual rates. This pattern is most pronounced in countries such as Egypt, Venezuela, Côte d'Ivoire, and Zimbabwe. Therefore, the puzzle is why households in these economies invest at levels far below what the standard model predicts as optimal.

2.2 Capital as a Public Good

The baseline model in Section 2.1 assumes capital investments are private for households. This means that households receive the full returns $f'(K)$ because they would own the invested capital in productions. These are scenarios where farmers invest in agricultural tools, vendors invest in mobile phones to gather market information, or freelance drivers invest in a motorcycle or car. However, capitals with public good features like infrastructure often require public investment commonly through government spending, and the benefits from these investments are shared among the population. Aschauer (1989) estimates a high average elasticity of 0.39 between output and public investment in the United States over 1949-1985, while Bom and Ligthart (2014) proposes a more

[1] $\frac{1}{\delta} - 1 \approx 0.04$, implying a 4% annual rate of time preference.
[2] Elasticity of intertemporal substitution of 0.5 is used in macro studies standard calibrations.
[3] Available for download at www.ggdc.net/pwt
[4] See https://www.rug.nl/ggdc/docs/the_next_generation_of_the_penn_world_table.pdf

Figure 1: Lower Capital Stock in Lower-Income Countries



Notes: Red line is OLS fit. Panel data from 1950 to 2023. Capital stock in millions US$ at 2021 PPPs. Output−side real GDP in millions US$ at 2021 chained PPPs. Source: PWT 11.0.

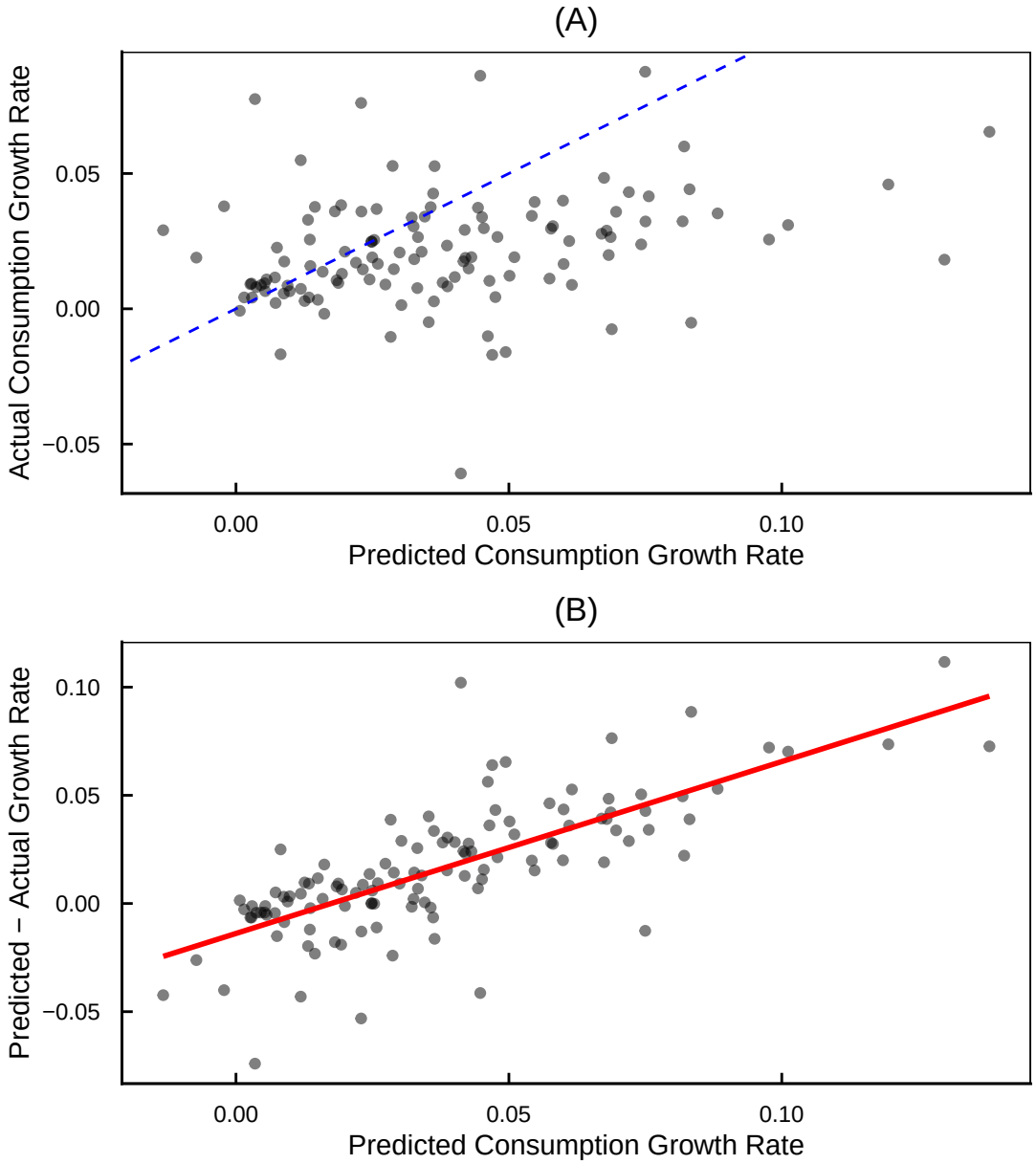
modest estimate of 0.106. The latter estimate that the short-run national-level elasticity of 0.083 still imply a 16% marginal return to public capital in OECD countries. The returns are likely higher in regional levels and much higher in lower-income countries with lower public capital stocks. However, since taxes commonly fund the majority of government spending, households engage with public investment indirectly through paying taxes. This structure can create a collective action problem and a government inefficiency problem that produces low levels of public investments despite high rates of return.

To illustrate, I will present an extreme case where we assume capitals are all publicly invested through taxation by governments. Consider the same optimisation problem as Equation 1 with a different budget constraint for homogeneous households in an economy. For any household i, the budget constraint is:

$$x_{i,t+1} = \frac{1}{N} \cdot \gamma f'(K_t) \cdot [(x_{i,t} - C_{i,t}) + \sum_{\forall j \neq i} (x_{j,t} - C_{j,t})] \tag{5}$$

where N is the total number of households in the economy, $\sum_{\forall j \neq i}(x_{j,t} - C_{j,t})$ is the aggregate public investment from all other households, and $\gamma \in [0, 1]$ represents government efficiency. Because public capitals can only be invested collectively, the total returns depend on the aggregate investment level, and assume each individual household receive an equal share of the total returns. Assume the government's only fiscal spending is on public capital investment, $\gamma = 0$ when the government effectively invests no tax revenue into public capitals and $\gamma = 1$ when the government effectively invests all of tax revenues into public capitals.

Figure 2: Actual vs. Predicted Average Consumption Growth Across Countries 2010-2019



Notes: The dashed line is the 45 degree line. The red line is OLS fit. Countries with populations below 1 million and irregular internal rates of return excluded. Data after 2019 is excluded to avoid Covid−19 distortions. Source: PWT 11.0.

Weak Taxation Capacity and Government Efficiency

To reflect weak taxation capacity, assume that household i makes public investment decision through taxation voluntarily and independent of other households' decisions. From the perspective of household i, the total returns from public capital investment as a public good is diluted among the population. Alternatively, for a fixed level of total returns on public capital, the impact of the household's investment level $(x_{i,t} - C_{i,t})$ on its assets in the next period is small for large N. Weak government efficiency also dilutes the return on public capitals when λ is small.

Solving for optimal intertemporal consumption path for household i given $\sum_{\forall j \neq i}(x_{jt} - C_{jt})$:

$$u'(C_{i,t}) = \frac{1}{N} \cdot \delta\gamma f'(K_t) u'(C_{i,t+1}) \tag{6}$$

Notice that when $N = 1$ and $\gamma = 1$, this equation is identical to Equation 3, but $u'(C_{i.t})$ decreases as N becomes larger and γ becomes smaller. For increasing and concave utility function, $u'(C_{i,t})$ is smaller when $C_{i,t}$ is larger. This means that household i will consume more and invest less as N increases and γ decreases. For sufficiently large N or sufficently small γ so that the Euler Equation cannot be satisfied by an interior solution, household i will optimally choose a corner solution of consuming all and invest none. $\frac{1}{N}$ produces a collective action problem where the global optimal decision is for all households to invest and benefit from high $f'(K)$ but individuals households have a dominant strategy of not investing. γ produces a problem where fiscal leakages to corruption or misallocation disincentives the household to invest.

This model captures the reluctant tax compliance in many low- or middle-income countries where the states have weaker taxation capacity and are relatively inefficient. Figure 3 also shows the model assumptions are consistent with the data.

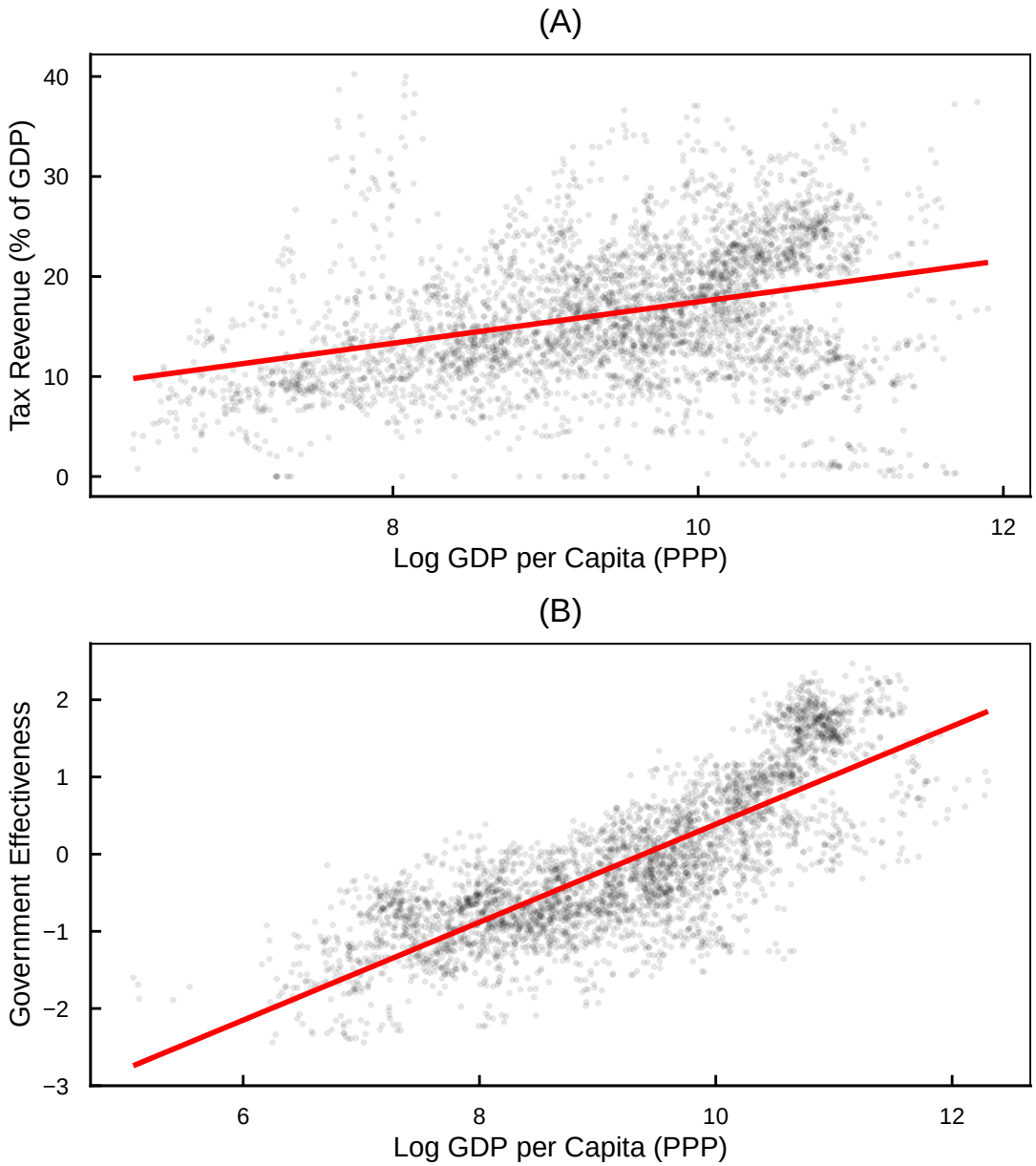
Strong Taxation Capacity and Government Efficiency

Higher income countries tend to have more efficient governments and stronger taxation capacity because income tax can be more easily collected from formal employees. Although, public investment is less voluntary for tax rates as given, households might make public investment decisions by influencing the tax rates through political channels. The key difference is that they expect other households to also comply with the decided upon tax rates.

Consider a simple case of homogeneous households facing the same optimisation problem as Equation 1 but assume instead perfect enforcability upon any chosen taxation level and perfect government efficiency. Any individual household i chooses a taxation level (public investment level) $(x_{i,t} - C_{it})$ and expect symmetry from other households. Because households are homogeneous, the following should hold:

$$[(x_{i,t} - C_{i,t}) + \sum_{\forall j \neq i}(x_{j,t} - C_{j,t})] = N \cdot (x_{i,t} - C_{i,t})$$

Figure 3: Higher Tax Revene and Government Effectiveness in Higher-Income Countries



Notes: Red line is OLS fit. Expenditure−side real GDP in millions US$ at 2021 chained PPPs. Government Effectiveness indicator from −2.5 to 2.5. Expenditure−side real GDP in millions US$ at 2021 chained PPPs. Sources: PWT 11.0, WDI

Since perfect government efficiency implies $\gamma = 1$, the budget constraint becomes identical to Equation 2:

$$x_{i,t+1} = \frac{1}{N} \cdot 1 \cdot f'(K_t) \cdot N \cdot (x_{i,t} - C_{i,t}) = f'(K_t)(x_{i,t} - C_{i,t})$$

Solving for optimal consumption path will produce an Euler Equation identical to Equation 3. Note that the collective action problem is resolved in this setting because the perfect tax enforcement enables households to optimise consumption conditional on the expectation of symmetrical behaviours from others.

A Generalised Model

In reality, the majority of countries likely fall in between the extreme cases outlined in Section 2.2. To generalise the model of public investment, I propose the following budget constraint:

$$x_{t+1} = \frac{1}{N^{\lambda}} \cdot \gamma f'(K_t^{pub})(x_t - C_t)$$

where K_t^{pub} is the public capital stock and λ is a parameter between 0 and 1 that determines the salience of the collective action problem in public investment. $\lambda = 1$ if the collective action problem is most extreme and $\lambda = 0$ if there is no collective action problem. Combining this model with the baseline private capital investment budget constraint gives:

$$x_{t+1} = r_t^{eff} \cdot (x_t - C_t) \tag{7}$$

where

$$r_t^{eff} = (1 - PublicShare_t) \cdot f'(K_t^{priv}) + PublicShare_t \cdot \frac{1}{N^{\lambda}} \cdot \gamma f'(K_t^{pub})$$

with K^{priv} denoting private capital stock and $PublicShare$ denoting the share of public capital investment in total investment. EXCLUDE:The returns on private and public capitals are derived by this generalised production function facing a representative household in the economy:

$$Y_t = A_t (K_t^{priv})^{\alpha^{priv}} (K_t^{pub})^{\alpha^{pub}} L^{1-\alpha^{priv}-\alpha^{pub}}$$

where the FOCs are:

$$f'(K_t^{priv}) = \alpha_{priv} \cdot \frac{Y_t}{K_t^{priv}} \quad \text{and} \quad f'(K_t^{pub}) = \alpha_{pub} \cdot \frac{Y_t}{K_t^{pub}} \tag{8}$$

Taken all together, solving for optimal consumption path under constraint given by Equation 7 produces an Euler Equation:

$$u'(C_t) = \delta \cdot r^{eff} \cdot u'(C_{t+1}) \tag{9}$$

and the consumption growth rate is given by:

$$\frac{C_{t+1}}{C_t} - 1 = (\delta \cdot r^{eff})^{\frac{1}{\sigma}} - 1 \tag{10}$$

3 Stylised Facts

A causal inference is outside the scope of this paper. However, in this section, I will highlight some stylised facts from data and literature on the relevant causal channels in support.

4 Roles of Collective Action and Government Efficiency in Development Accounting

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